



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to S.R.P. AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

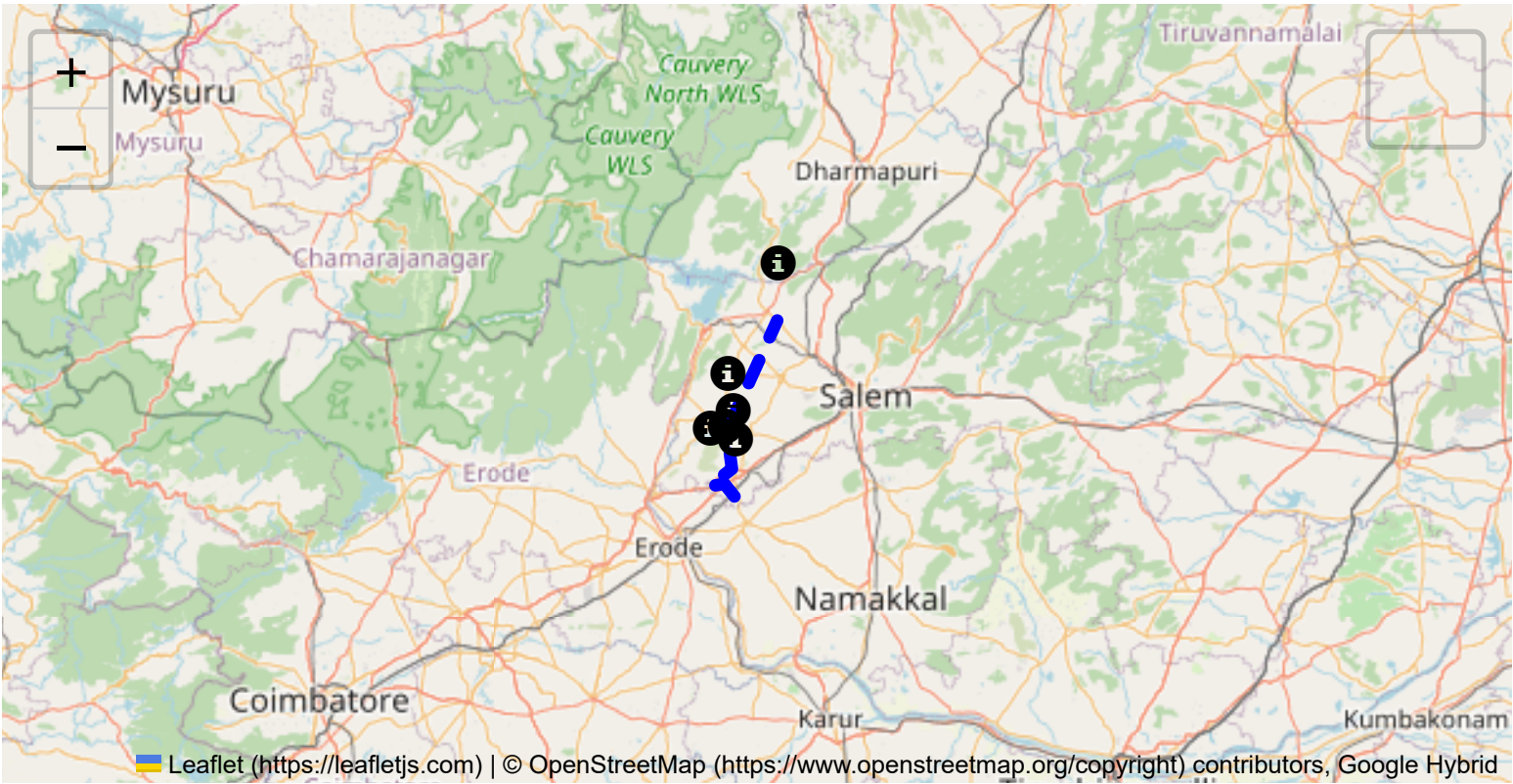
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 63.53 km
Estimated Duration: 1.5 hours
Adjusted Duration (Heavy Vehicle): 1.9 hours
Start: (11.4381, 77.8734)
End: (11.81331055, 77.96554821)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to RX78+86M, Salem covers approximately 63.53 kilometers. It primarily follows Bhavani Main Road through Sankari West and Padaiveedu and includes stretches on Sankari - Edappadi - Mecheri Rd. The journey typically takes about 1.5 hours for heavy vehicles transporting hazardous materials.

Weather Conditions

1. **Typical Weather:** The region experiences a tropical climate. Summers (March to June) are hot with temperatures up to 40°C, while monsoons (July to September) are characterized by heavy rainfall. Winters (December to February) are mild.
2. **Weather-Related Hazards:** Heavy rains during the monsoon may lead to waterlogging and slippery roads, which can pose a risk for heavy vehicles. Additionally, fog in early winter mornings might affect visibility.

Traffic Patterns

1. **Peak Hours:** Congestion is common during morning (8:00 AM - 10:00 AM) and evening (5:00 PM - 7:00 PM) peaks, especially near towns like Sankari and Padaiveedu.
2. **Congestion-Prone Areas:** Please be cautious when passing through market areas and major intersections in these towns. Traffic may also delay progression near schools during start and end times.

Road Quality and Infrastructure

- **Assessment:** The main road segments like Bhavani Main Road are generally in good condition, but some internal roads may have uneven surfaces and potholes, especially after monsoons.
- **Infrastructure:** Signage is generally clear, but additional caution is advisable in less illuminated rural areas at night.

Alternative Routes

In case of emergencies or traffic delays, consider using NH44, which runs parallel to sections of the route. This highway is well-maintained but be aware of entry and exit ramps.

Local Regulations

1. **Hazardous Materials Transport:** Transport of hazardous materials requires proper documentation and adherence to the rules outlined by the Central Motor Vehicles Act. Restrictions may apply in populated areas and near schools.
2. **Permits:** Ensure all necessary permits and documentation for transporting hazardous materials are in place.

Historical Incidents

- While comprehensive data on incidents may not be available publicly, reports of accidents involving heavy vehicles are occasionally noted in areas with winding roads or following adverse weather conditions. Vigilance in these regions is recommended.

Environmental Considerations

- **Sensitive Areas:** Avoid operations during peak agricultural activities. Areas near water bodies require extra caution as spills and contaminants can lead to severe ecological damage.
- **Ecosystems:** Be mindful of wildlife crossings in rural segments, particularly during night travel.

Communication Coverage

- **Coverage Analysis:** Mobile network coverage is generally robust in urban and semi-urban segments. However, expect occasional dead zones in more rural stretches. A satellite phone or radio could be a beneficial backup.

Estimated Emergency Response Times

- **Urban Segments:** Emergency response times in urban areas like Sankari are moderate (15-30 minutes).
- **Rural Areas:** In rural stretches, response times might extend to 45 minutes to an hour, depending on the incident's location.

Overall Risk Assessment Summary

The route presents moderate risks typical for regional roads dealing with varied weather conditions, predictable traffic congestion, and occasional infrastructure limits. For heavy vehicle operators transporting hazardous materials:

- **Plan around peak traffic times** to minimize delays and exposure.
- **Monitor weather forecasts** to avoid precipitation hazards.
- **Maintain consistent communication** with logistics and emergency services, especially in rural segments.
- **Stay informed about local regulations** surrounding hazardous material transport.

This careful preparation will help ensure a safer journey for drivers and minimise potential risks associated with the transport of hazardous materials on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44900, 77.87406	30 KM/Hr
5	Turn	Medium	11.45358, 77.85781	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
6	Turn	High	11.45353, 77.85697	15 KM/Hr
7	Turn	Medium	11.45853, 77.85139	30 KM/Hr
8	Turn	Medium	11.45906, 77.85132	30 KM/Hr
9	Turn	Medium	11.46307, 77.84941	30 KM/Hr
10	Turn	Medium	11.46316, 77.84921	30 KM/Hr
11	Turn	Medium	11.45543, 77.81740	30 KM/Hr
12	Turn	Medium	11.45563, 77.81708	30 KM/Hr
13	Turn	High	11.45634, 77.81673	15 KM/Hr
14	Blind Spot	Blind Spot	11.49390, 77.86748	10 KM/Hr
15	Turn	Medium	11.49513, 77.86656	30 KM/Hr
16	Turn	Medium	11.50866, 77.86053	30 KM/Hr
17	Turn	Medium	11.54750, 77.85273	30 KM/Hr
18	Turn	Medium	11.55039, 77.85511	30 KM/Hr
19	Turn	Medium	11.55481, 77.85639	30 KM/Hr
20	Turn	High	11.57893, 77.85430	15 KM/Hr
21	Turn	Medium	11.57903, 77.85441	30 KM/Hr
22	Turn	Medium	11.58012, 77.85736	30 KM/Hr
23	Turn	High	11.58755, 77.85817	15 KM/Hr
24	Turn	High	11.59148, 77.86400	15 KM/Hr
25	Turn	Medium	11.59294, 77.88170	30 KM/Hr
26	Turn	High	11.59306, 77.90633	15 KM/Hr
27	Blind Spot	Blind Spot	11.59207, 77.90646	10 KM/Hr
28	Turn	Medium	11.63857, 77.94831	30 KM/Hr
29	Turn	Medium	11.64207, 77.95199	30 KM/Hr
30	Blind Spot	Blind Spot	11.69590, 77.97114	10 KM/Hr
31	Turn	High	11.70129, 77.96360	15 KM/Hr
32	Turn	Medium	11.70140, 77.96366	30 KM/Hr
33	Turn	Medium	11.73438, 77.96729	30 KM/Hr
34	Turn	High	11.77777, 77.99630	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
2	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
3	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
4	hospital	Veda Maruthuvamanai Hospital	11.6956155, 77.9678122	30 km/h	Medium
5	clinic	E.V.P Vairavel Clinic	11.6954975, 77.9667903	30 km/h	Medium
6	clinic	AMS Clinic	11.752664, 77.974974	30 km/h	Medium
7	hospital	Tholasampatti, Government Hospital	11.75638, 77.973813	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45358, 77.85781



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45853, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45906, 77.85132



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45543, 77.81740



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45563, 77.81708



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45634, 77.81673



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49390, 77.86748



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49513, 77.86656



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50866, 77.86053



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.54750, 77.85273



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55039, 77.85511



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55481, 77.85639



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57893, 77.85430



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57903, 77.85441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58012, 77.85736



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58755, 77.85817



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59148, 77.86400



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59294, 77.88170



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59306, 77.90633



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.59207, 77.90646



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63857, 77.94831



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64207, 77.95199



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.69590, 77.97114



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70129, 77.96360



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70140, 77.96366



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73438, 77.96729



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77777, 77.99630

